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Thesis

A PICTURE COMPRESSES TEXT

- Long text is expensive for LLMs because attention grows quadratically with sequence length.
- Rendering text into a 2D image can encode far more content per visual token than digital text tokens.
- OCR becomes a measurable compression-decompression testbed and a route toward visual long-term memory.

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Architecture

DEEPEncoder + MOE DECODER

- DeepEncoder serially combines an 80M SAM window encoder, 16x convolutional compression and 300M CLIP global attention.
- A 1024 x 1024 image begins with 4,096 patches but enters global attention as only 256 vision tokens.
- DeepSeek-3B-MoE activates about 570M parameters to decode the compressed representation back to text.

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Resolution & data

64 TO 1,853 TOKENS

- Tiny/Small/Base/Large output 64/100/256/400 tokens; Gundam modes tile extreme-resolution pages.
- Training is 70% OCR, 20% general vision and 10% text-only; OCR 2.0 covers charts, chemistry and geometry.
- First train DeepEncoder, then the end-to-end VLM with the tokenizer portion frozen.

DEEPSEEK-OCR

DeepSeek-OCR

Contexts Optical Compression | 2025

Treat a rendered document image as compressed text: a high-resolution vision encoder converts hundreds or thousands of text tokens into a small visual-token set that an MoE decoder reconstructs.

OCR

context compression

vision tokens

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Compression

QUALITY VS FORGETTING

- Below 10x text-to-vision compression, OCR precision is about 97%; 10-12x retains about 90%.
- At nearly 20x, precision remains around 60%, revealing a controllable lossy-compression regime.
- Older dialogue could be rendered and progressively downsampled, mimicking biological forgetting.

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Practical evidence

FEWER TOKENS, STRONG OCR

- With 100 tokens it beats GOT-OCR2.0 using 256; with under 800 it beats MinerU2.0 using nearly 7,000.
- The unified interface handles documents, charts, formulas, geometry and natural images.
- Throughput exceeds 200k pages/day on one A100-40G; 20 x 8-GPU nodes can produce 33M pages/day.

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Meaning & limits

A PROOF OF CONCEPT

- Visual compression may keep recent context high-fidelity while old context fades efficiently.
- OCR alone does not prove reliable reasoning over compressed memories or needle retrieval.
- Future work needs interleaved optical-text pretraining, retrieval tests and deeper distortion studies.